Can AI Scientists Correct a Wrong Prior? Controlled Experiments in Executable Chemical Worlds

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Abstract

Scientific agents rarely begin from a blank state: they inherit model knowledge, task descriptions and investigator-supplied assumptions. A useful endpoint therefore does not show whether an agent discovered the governing relation, followed a correct prior or succeeded despite a wrong one. We formulate prior use and correction as controlled experimental variables in executable chemical worlds. The same hidden world, public operations, evidence budget and safety surface are paired across three conditions: anonymous materials without a task-specific dossier, directionally useful nominal information and an equally detailed but deliberately misindexed dossier. Each agent controls a multi-experiment campaign through one persistent operation-level session, records predictions and belief updates at fixed checkpoints, and commits an executable law summary and final recommendation. Evaluator-owned counterfactual queries and blind replay separate epistemic recovery from endpoint optimization. A completed DeepSeek development matrix spanning five tasks, three prior arms and five world seeds per task reached terminal records for all 75 scheduled cells while retaining failures, resource rejections and tool-recovery tails. The matrix shows that explicit dossiers can substantially change experimental behavior and best endpoints, while verbal warnings do not selectively distinguish aligned from misindexed information. These observations motivate a preregistered five-task study whose primary estimand tests whether evidence reduces held-out prediction error more strongly under a wrong prior than under a correct prior without degrading the correct-prior condition. The resulting framework distinguishes prior confirmation, evidence-driven correction, local policy repair and reusable law discovery, and provides a process-complete basis for studying when an AI scientist can abandon a plausible but incorrect scientific belief.

1. Introduction

Scientific discovery is not simply the production of a high-scoring outcome. A researcher can obtain a useful result for several different reasons: a prior model was already correct, an incorrect model was repaired by evidence, a local heuristic happened to work, or the endpoint was reached without a reusable account of the underlying relation. These explanations are scientifically distinct even when their final scores are identical.

This distinction is increasingly important for AI systems that choose and execute experiments. Language-model agents can plan syntheses, call chemistry tools, operate instruments and participate in self-driving laboratory workflows [4, 5, 7, 15–17]. Interactive scientific environments likewise test whether agents can formulate hypotheses, acquire evidence and recover hidden rules [1, 8, 10, 12, 19, 20]. Yet an agent's apparent scientific success can remain difficult to interpret because pretrained knowledge, prompt-provided information, experiment selection, endpoint optimization and verbal explanation are usually entangled.

The central problem is the prior. An agent entering an experimental campaign may receive no additional task-specific information, a useful but incomplete description, or a plausible description attached to the wrong material. The important capability is not merely whether the prior changes behavior. A scientifically adaptive agent should decide which evidence is worth acquiring, use observations to revise its predictions, reduce confidence in a contradicted prior, summarize the recovered relation in an executable form and transfer that relation to conditions it has not directly tested. Conversely, an agent may preserve an incorrect prior through selective measurement, reinterpret contradictory observations or patch only its action policy.

Physical self-driving laboratories are indispensable for real-material validation, but they make this causal question difficult to study at scale. Strictly matching hidden laws, material identities, measurement noise, budgets and safety conditions across alternative prior states is expensive and often impossible. ChemWorld provides a complementary experimental instrument: chemical processes, observations and private material laws can be instantiated as executable worlds, while the public experimental interface and complete operation history remain controlled. This permits the prior condition to change while the hidden world and evidence opportunity remain fixed.

Here we introduce a controlled framework for studying prior use, wrong-prior correction and law transfer in experimental agents. It makes four contributions.

1. **Prior quality becomes a matched intervention.** Anonymous, aligned and misindexed information conditions share the same world, information volume, public contract and resource card.
2. **Discovery is evaluated through predictions and**

actions, not self-report alone. Fixed checkpoints bind beliefs to evaluator-owned counterfactual queries and to the next experimental operation selected by the agent.

3. **Endpoint success is separated from reusable understanding.** Blind outcome replay, executable law summaries and transfer tests distinguish local optimization from law recovery.
4. **The complete experimental process remains auditable.** One persistent session controls multiple experiments under a shared resource ledger, while failures, invalid actions, stopping and exact replay remain part of the outcome rather than being silently repaired.

Development results already establish a consequential boundary: explicit priors reshape behavior, but nominal-information warnings and endpoint gains do not reliably reveal whether the supplied prior is correct. The formal study therefore asks a narrower and harder question: does autonomously acquired evidence selectively repair the wrong prior?

2. Related work

2.1 Autonomous experimentation and laboratory agents

Autonomous chemistry systems combine planning, analysis and physical execution. Coscientist, ChemCrow, A-Lab and instrument-facing agents demonstrate tool use, synthesis planning, materials search and closed-loop experimentation on real equipment [4–7, 15, 16]. These systems establish that AI agents can participate in consequential experimental workflows. Their main evidential strength is physical validity; their practical constraint for the present question is the cost of repeatedly constructing matched alternative worlds in which only the correctness of a prior changes.

2.2 Virtual chemistry and process environments

Summit and Olympus support repeatable reaction optimization and experimental-design benchmarks, while PC-Gym exposes nonlinear process-control problems [3, 9, 11]. ChemGymRL provides modular interactive benches for reaction, extraction, distillation and characterization, enabling reinforcement-learning agents to act within a virtual chemistry laboratory [2]. MADE extends closed-loop discovery benchmarks to budget-constrained materials settings [13]. These systems make experimentation cheaper, safer and more repeatable than physical execution, but they are generally used to compare policies or optimize endpoints under one task definition. The present study instead treats the correctness of agent-facing scientific information as the experimental intervention.

2.3 Interactive scientific-discovery benchmarks

DiscoveryWorld, BoxingGym, SciGym, SciExplorer and NewtonBench organize tasks around repeated cycles of hypothesis formation, intervention and inference [8, 10, 12, 14, 20]. CausaLab and ReplaySCM further emphasize interventional causal discovery and executable mechanism induction [1, 19]. SciDisco uses process-verifiable discovery environments to provide intermediate training signals [18]. These studies motivate evaluating the trajectory of discovery rather than only the final answer. Our focus is complementary: the hidden law does not change during a campaign; what changes is whether an explicit prior is absent, useful or systematically wrong.

2.4 The unresolved identification problem

Three ambiguities remain when endpoint score, belief statement and scientific understanding are not separated. First, a correct prior can make an agent look like a rapid discoverer even if it merely confirms supplied information. Second, a wrong prior can improve an endpoint by encouraging useful exploration without ever being rejected. Third, a verbal law summary can be correct while the agent's subsequent predictions or actions remain inconsistent with it. A matched prior intervention, typed checkpoints and evaluator-owned tests are needed to distinguish these cases.

3. Conceptual framework

3.1 Fixed world, variable prior

Each comparison begins with one executable world containing a fixed evaluator-owned law. The public task, action space, observation channels, resource card, safety rules and bound stochastic identity are held constant. Only the task-specific dossier changes:

- **Opaque:** materials use anonymous identifiers and no additional task-specific dossier.
- **Aligned nominal:** each anonymous identifier is paired with incomplete nominal properties that are directionally useful in the fixed world.
- **Misindexed nominal:** the same property bundles, fields, values, wording and confidence language are retained, but the bundles are permuted across material identifiers.

Experimental evidence is explicitly authoritative in all conditions. The misindexed arm is therefore not a trick question about obedience; it tests whether the agent seeks and uses evidence that can override a plausible prior.

3.2 Operation, experiment and campaign

An operation is one submitted physical or measurement action. A complete experiment begins with a new batch and ends with a committed final assay or an allowed discard. A discovery campaign comprises multiple complete experiments under one fixed world, one persistent agent session and one shared resource ledger. Physical batch state resets between experiments, but the public history, agent context, hidden law and remaining resources persist.

This distinction matters because repeated operations and experiments inside one campaign are nested observations, not independent scientific samples. The independent analysis unit is the matched task-by-world cluster.

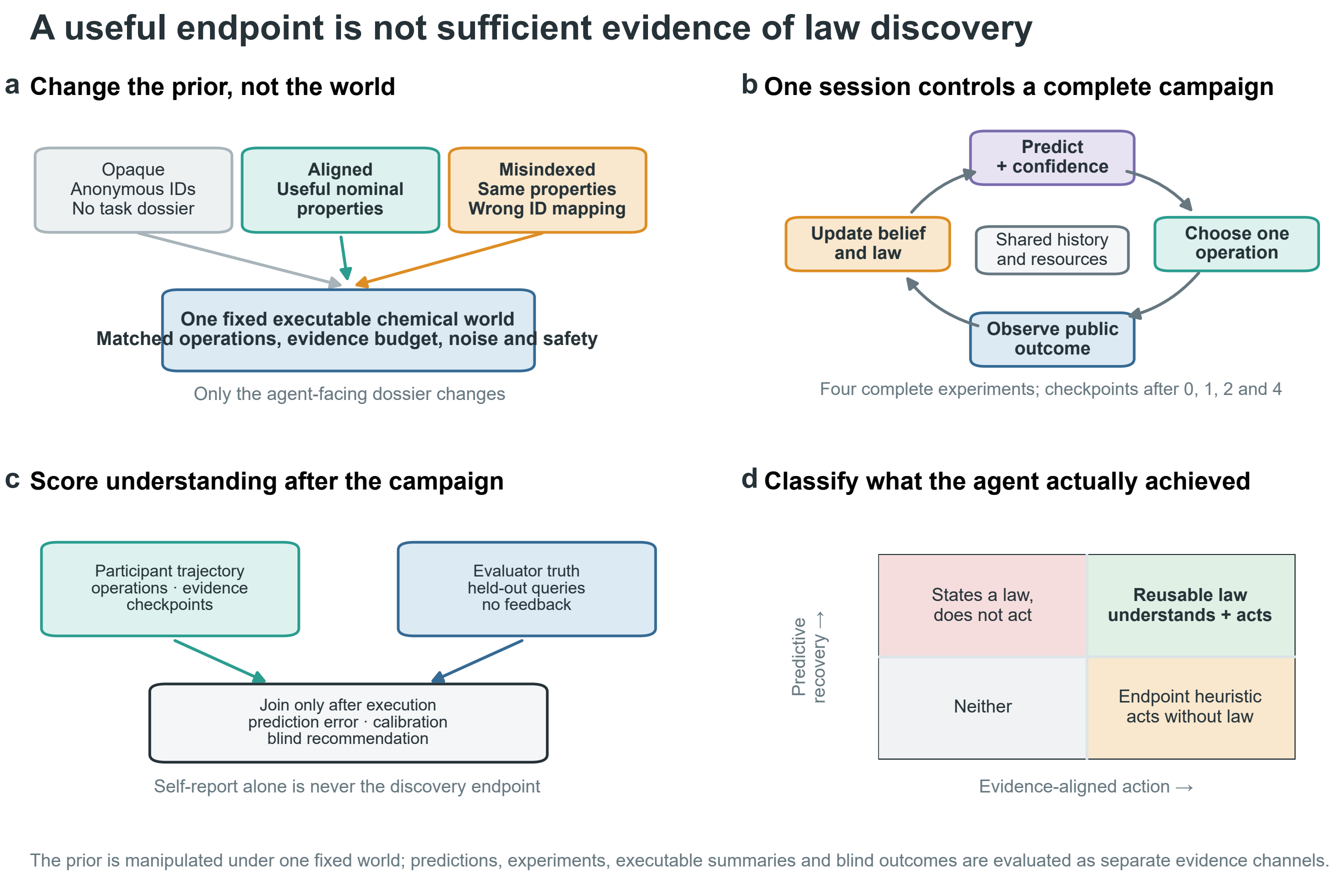

Figure 1. From prior to reusable law. a, Opaque, aligned and misindexed dossiers enter the same fixed executable world with matched operations, evidence budget, noise and safety surface. **b,** One persistent session repeatedly predicts, selects an operation, observes the public outcome and updates its belief and executable law summary across a shared-resource campaign. **c,** Participant trajectories and evaluator-owned held-out truth remain separate until the campaign reaches a terminal state; prediction error, calibration and blind recommendation outcomes are scored afterward. **d,** Predictive recovery and evidence-aligned action define four distinguishable phenotypes. Only their joint success, followed by transfer, supports a reusable-law claim; endpoint success or a correct statement alone does not.

3.3 Four separable outcomes

We distinguish four outcomes that are often collapsed into a single score.

1. **Endpoint optimization:** whether the campaign identifies a high-quality experimental outcome.
2. **Predictive recovery:** whether held-out counterfactual prediction error decreases.
3. **Prior correction:** whether evidence selectively improves the wrong-prior condition without degrading the correct-prior condition.
4. **Reusable law recovery:** whether the final executable summary predicts unseen continuous conditions and supports held-out control or transfer.

An agent can succeed on any subset. In particular, endpoint success without predictive and transfer validity is classified as local optimization rather than law discovery.

4. Study design

4.1 Chemical-world cohort

The formal public cohort spans five task families: electrochemical conversion, reaction followed by crystallization, reaction followed by distillation, phase-partition discovery and safety-constrained reaction. Five independently selected public worlds per task yield 25 task-by-world clusters. Each cluster contains the three prior arms, for 75 participant cells. Development and method-qualification worlds are disjoint from this cohort. A separate private cohort is committed in advance and remains sealed until public analysis is frozen.

Environment qualification and participant outcomes form separate evidence layers. Environment tests establish that the hidden relations are coherent, identifiable and executable through the public measurement surface. They do not show that an agent discovers those relations.

4.2 Persistent experimental agent

Each cell is controlled by one persistent Codex process and one provider session. After every public outcome, the participant chooses the next operation through the host-owned laboratory tool. The host validates schemas, executes transactions, updates resources and protects private state, but does not select or repair scientific actions.

Each campaign contains four complete experiments and typed checkpoints before evidence, after the first and second experiments, and at the end of the fourth experiment. A checkpoint records the agent's assessment of prior reliability, predictions, uncertainty, evidence references, executable law summary and next experimental intent. Checkpoints do not create additional provider sessions.

4.3 Evaluator-owned evidence

Four registered counterfactual queries per task-world cluster are executed independently by the evaluator. Their truth is shared across prior arms and checkpoints and is never returned to the participant. The primary prediction error is the mean normalized absolute error across registered query-metric pairs.

After the campaign, the participant commits one completed experiment as its final recommendation. The evaluator then performs paired blind replay of the observed incumbent and the committed recommendation. These executions use separate resources and do not enter participant-operation or provider denominators.

4.4 Hypotheses and estimands

Let $E_{a,k}$ denote held-out prediction error for prior arm a at checkpoint k. The primary hypothesis tests selective evidence-driven correction:

$$\begin{aligned} C_{\text{prior}} = (&E_{\text{misindexed,pre}} - E_{\text{misindexed,final}}) \\ &- (E_{\text{aligned,pre}} - E_{\text{aligned,final}}). \end{aligned}$$

Success requires the lower confidence bound for C_{prior} to exceed zero, the wrong-prior condition to improve, and the aligned condition not to deteriorate beyond a prespecified tolerance. Correct-prior utility, wrong-prior vulnerability and knowledge-to-action translation form a hierarchical secondary family. Endpoint, calibration, behavior, law-summary, transfer, resource and safety outcomes are reported as separate channels rather than one leaderboard score.

Failed scientific cells remain in the denominator and are not replaced. A right-censored cell carries its last valid checkpoint forward; a missing final prediction receives zero primary improvement. Only a pure infrastructure failure without persisted trajectory may resume under the frozen attempt cap.

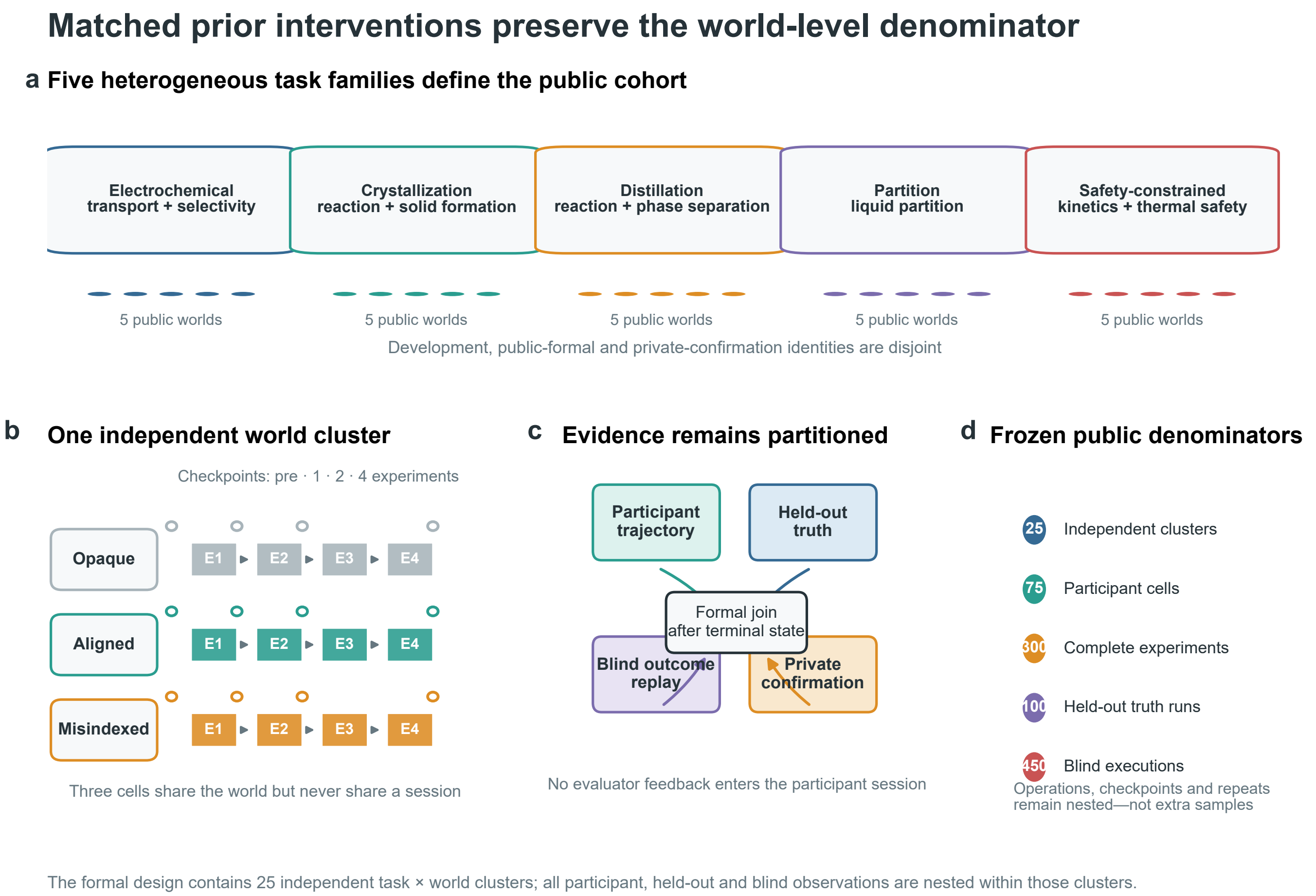

Figure 2. Frozen cohort and evidence architecture. a, Five task families each contribute five public-formal worlds selected independently of participant outcomes; development, public and private identities are disjoint. **b,** Each independent task–world cluster contains opaque, aligned and misindexed participant cells. Every cell uses one persistent session for four complete experiments, with checkpoints before evidence and after one, two and four experiments. **c,** Participant trajectories, evaluator-held truth, blind outcome replay and private confirmation retain separate resources and records and are joined only after terminal execution. **d,** The planned public denominator comprises 25 independent clusters, 75 participant cells and 300 complete experiments, with 100 provider-free held-out truth executions and 450 provider-free blind executions nested within the corresponding clusters. These are frozen design denominators, not completed outcomes.

5. Development evidence

The following results qualify the method and sharpen the scientific question. They are not part of the public formal or private-confirmation denominator, and the two provider configurations are not used for a cross-provider capability ranking.

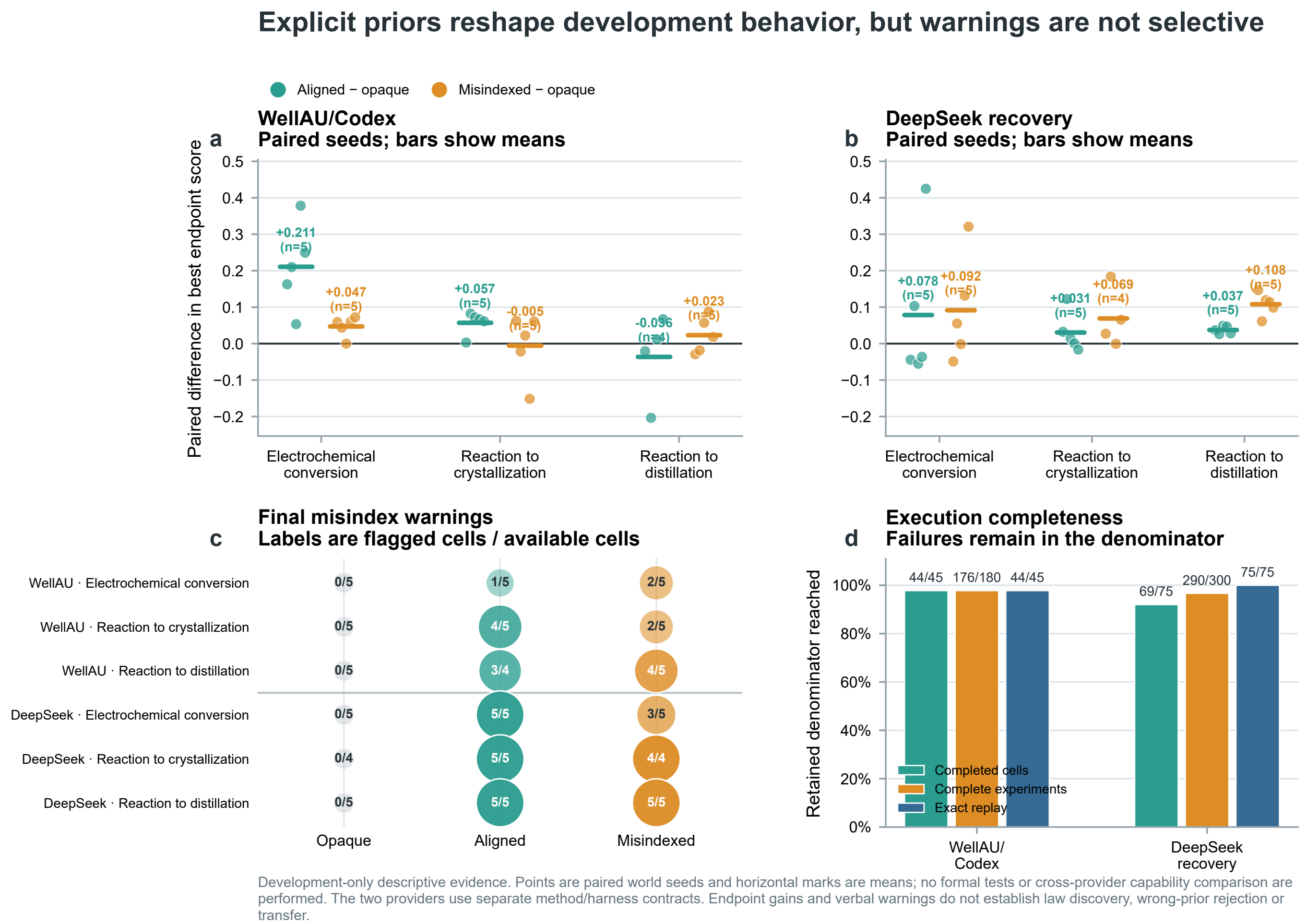

Figure 3. Provider-separated development evidence for prior-sensitive behavior. a,b, Paired world-seed differences in the best endpoint observed during four-experiment campaigns for aligned versus opaque and misindexed versus opaque information. Points are retained paired seeds and horizontal bars are descriptive means. WellAU/Codex contains five pairs per task except the aligned distillation contrast ($n = 4$); DeepSeek recovery contains five pairs except the misindexed crystallization contrast ($n = 4$). **c,** Final explicit misindex warnings, shown as flagged cells over available terminal belief records for every provider, task and prior arm. **d,** Completed-cell, complete-experiment and exact-replay denominators. Failures remain in the scheduled or terminal denominator and are not replaced. Panels a–c retain the common three-task paired endpoint/warning source used for provider-separated continuity; the complete five-task DeepSeek operational denominator is shown in the closeout table. All panels are development-only descriptive summaries; no confidence interval, formal hypothesis test or cross-provider capability comparison is performed. Endpoint gains and verbal warnings do not establish law discovery, selective wrong-prior correction or transfer.

5.1 Explicit priors reshape endpoint behavior

One development matrix used the frozen persistent-session interface with a WellAU-provided Codex model. It produced 44 completed cells out of 45 and 176 complete experiments out of 180. Mean paired aligned-minus-opaque differences in the best observed endpoint were +0.211 for electrochemical conversion, +0.057 for crystallization and -0.036 for distillation, with the distillation contrast based on four complete pairs. Misindexed information was not consistently harmful, and explicit misindex warnings included substantial false positives in aligned cells.

A recovery-amended DeepSeek development block was completed across all five task families, retaining the immutable seed-0 gate records for partition discovery and safety-constrained reaction and adding their seeds 1–4 in a separate continuation block. No seed-0 outcome was rerun or replaced. Every scheduled cell reached a terminal record (**75/75**); **69/75** cells were completed and qualified, with **290/300** complete experiments, **2,663/2,587** operation attempts/committed operations, **73** validation failures, **3** resource rejections, **69** recovered MCP failures and **0** provider-error events. Exact physical/resource replay passed for **75/75** terminal trajectories. Provider accounting covered 72/75 cells; the remaining three stopped before a provider terminal event and retain usage as unavailable rather than zero. The complete development block is still descriptive: it has no evaluator-truth prediction scoring, blind recommendation replay, private transfer confirmation or formal hypothesis test, and provider groups are never pooled into a capability ranking.

The ordering is not aligned, opaque, then misindexed. In distillation, every paired seed favored both explicit-

Table 1. Five-task DeepSeek development closeout. All five task families reached the scheduled five-seed terminal denominator. The partition and safety continuation preserved their immutable seed-0 gate outcomes; their operational rows are reported descriptively and are not pooled with the common three-task paired endpoint panels.

Task	Terminal	Qualified	Experiments	Attempts/committed	MCP failures	Exact replay
Electrochemical conversion	15/15	15/15	60/60	373/372	13	15/15
Reaction to crystallization	15/15	13/15	54/60	606/605	18	15/15
Reaction to distillation	15/15	15/15	60/60	637/637	7	15/15
Partition discovery	15/15	12/15	58/60	553/501	14	15/15
Safety-constrained reaction	15/15	14/15	58/60	494/472	17	15/15

Table 2. DeepSeek five-task development endpoint contrasts. Values are descriptive paired-seed means in best endpoint score; the partition and safety rows are continuation observations and are not pooled into the common three-task Figure 3 endpoint panels.

Task	Aligned–opaque	Misindexed–opaque
Electrochemical	$+0.0785$ ($n = 5$)	$+0.0915$ ($n = 5$)
Crystallization	$+0.0305$ ($n = 5$)	$+0.0690$ ($n = 4$)
Distillation	$+0.0374$ ($n = 5$)	$+0.1080$ ($n = 5$)
Partition	-0.1397 ($n = 5$)	-0.1063 ($n = 5$)
Safety-constrained reaction	$+0.0223$ ($n = 5$)	-0.0196 ($n = 5$)

information arms over opaque identifiers, and the misindexed mean gain was larger. In partition discovery, both explicit-information contrasts were negative, whereas the safety task was mixed. These task-pattern differences are descriptive and are partly coupled to the continuation contract; they do not support a pooled provider effect. A better endpoint therefore cannot be treated as evidence that the agent accepted a correct prior, rejected a wrong prior or recovered the hidden law.

5.2 Verbal suspicion is not selective correction

Across available terminal belief records, final DeepSeek misindex warnings were 0/5, 5/5 and 3/5 for opaque, aligned and misindexed electrochemical cells; 0/4, 5/5 and 4/4 in crystallization; 0/5, 5/5 and 5/5 in distillation; 0/4, 2/4 and 3/4 in partition discovery; and 0/5, 2/4 and 0/5 in safety-constrained reaction. The model therefore often associated dossier presence with possible misindexing, but the pattern was task-dependent and did not selectively distinguish the correct and incorrect dossiers. Mean aligned-minus-misindexed changes in self-reported prior reliability were small or heterogeneous across tasks, including negative changes in several continuation cells.

This is a scientifically useful negative boundary. A warning token or reduced stated confidence is not a valid bias-rejection endpoint unless it predicts evaluator-scored correction and subsequent evidence-aligned action.

5.3 Persistent-session accounting exposes a separate operational layer

The 75 terminal DeepSeek trajectories accumulated **267,929,149 input tokens**, including **260,033,536 cached tokens** (97.05%), **7,895,613 uncached input tokens** and **2,932,468 output tokens**. These are cumulative turn-level provider counters for long-lived sessions. The large cached fraction therefore reflects reuse of the shared prompt and growing campaign history; it is not repeated model output and does not represent additional independent experimental evidence. Usage was reconciled for 72/75 cells. The other three cells stopped before a provider terminal event, so their usage remains unavailable rather than being imputed as zero.

Operational failures also require their own denominator. The matrix recorded 73 schema-validation failures and 69 recovered MCP tool failures but no provider-error events. Moreover, 76 of 2,663 operation attempts did not become committed operations. Thus, a zero provider-error count would have hidden most of the execution burden: model-to-tool conformance, recovery and resource preflight were more consequential than provider transport in this block. These events are properties of the complete agent system and remain separate from both physical outcomes and independent world-level scientific samples.

5.4 Development conclusion

Across the retained development configurations, explicit priors clearly alter the course and endpoint of experimentation. The same evidence does not show selective wrong-prior rejection. It instead demonstrates why the formal study must score held-out predictions, executable summaries and blind recommendations rather than infer understanding from endpoint performance or verbal self-report.

6. Formal results

This section will report the preregistered public matrix only after every scheduled cell has reached a completed, right-censored or failed terminal state. Arm contrasts will remain blinded during execution. The section is reserved for:

- the primary selective-correction contrast and its three required conditions;
- correct-prior utility and wrong-prior vulnerability;
- checkpoint-level prediction-error and calibration trajectories;
- executable law-summary validity and continuous counterfactual prediction;
- evidence-seeking, selective measurement, attribution and prior-persistence phenotypes;
- blind incumbent-versus-recommendation outcomes;
- complete failure, censoring, resource, token, cost and wall-time denominators.

No development value will be substituted for a missing formal result.

7. Private confirmation and transfer

After the public method and analysis code are frozen, the committed private cohort will be executed once. The same prediction, law-summary and control metrics will test world-held-out and mechanism-family transfer. Private identities will not be revealed to the participant, and a negative or incomplete result will not trigger a replacement run.

The strongest claim—discovery of a reusable law—requires joint evidence: selective wrong-prior correction, aligned-prior non-degradation, an executable typed law summary, valid predictions on unseen continuous conditions and successful preregistered transfer. Anything less will be reported as confirmation, local correction, policy repair or endpoint optimization according to the observed combination.

8. Discussion

8.1 Why endpoint success is insufficient

The development matrices show that a deliberately wrong dossier can improve the best observed endpoint. This is not paradoxical. A prior can change where an agent searches, increase experimental diversity or direct attention toward a useful region even when its material mapping is wrong. Endpoint improvement measures utility of the induced search trajectory, not truth of the agent's scientific model.

8.2 Bias rejection must be behaviorally selective

The broad tendency to distrust any explicit dossier illustrates a second ambiguity. Generic skepticism can generate the right verbal stance without identifying the wrong prior. A valid correction measure must be selective: contradictory evidence should improve wrong-prior predictions more than correct-prior predictions, and that improvement should influence subsequent experimental choices.

8.3 Scientific understanding as a coupled capability

Prediction, explanation and action can dissociate. An agent may understand a relation but lack the operational competence or remaining resources to exploit it; it may act successfully without an accurate model; or it may state the right model while continuing to choose inconsistent experiments. The joint evaluation therefore supports four interpretable phenotypes: understands and acts, understands but cannot act, acts without understanding, and neither.

8.4 The harness is part of the evaluated agent system

A persistent Codex session contributes capabilities that a stateless model call does not: it retains the campaign history, chooses among tools after each observation and can revise a plan without a host reconstructing its reasoning state. The same harness also introduces failure surfaces through tool discovery, schema conformance, retry rules, context growth and checkpoint submission. Provider-side caching changes resource use but not the number of experiments or independent worlds. Consequently, the participant is the frozen combination of model, reasoning setting, prompt, Codex runtime, MCP interface and resource policy—not the model weights in isolation. A cross-model claim requires these components to be matched or explicitly manipulated; the present development providers are therefore reported separately.

8.5 Scope and limitations

The study evaluates bounded executable chemical worlds rather than universal chemical fidelity or direct wet-laboratory validity. A single frozen participant method supports conclusions about that agent-system configuration, not language models in general. Five task families provide mechanistic breadth but only five independent public worlds per task, so the confirmatory design is powered for moderate-to-large effects rather than subtle differences. Evaluator queries are registered before participant execution, but any finite query set samples only part of a hidden law. Finally, exact software replay does not eliminate model-provider variability; provider attempts and session failures are reported as operational characteristics rather than independent scientific samples.

9. Methods

9.1 World and prior construction

Each task instantiates a public experimental contract and a private evaluator-owned material or process law. Public formal worlds are selected deterministically from a namespace disjoint from development worlds. Within each world cluster, the three prior arms share the hidden law, public contract, resource card and stochastic identity. Aligned and misindexed dossiers contain identical fields, values, wording and confidence language; the latter applies a frozen permutation to the material identifiers.

9.2 Transactional execution and resources

Every operation enters schema validation and resource preflight before candidate execution. Committed operations update physical state and the campaign ledger; invalid or resource-rejected attempts retain their declared reporting debit without entering committed physical state. Task-specific resource cards bound vessel starts, assays, measurements, stocks, process time, repeated operations, quench and transfer time, and closeout reserve across all experiments in a campaign.

9.3 Persistent Codex/MCP execution

Each participant cell launches one Codex Responses process and retains one provider session across the four complete experiments. Web search is disabled. The participant instructions prohibit shell use, file changes and repository inspection and require physical decisions to pass through the host-owned `chemworld_lab` STDIO MCP server. The bounded domain tools expose material information, belief checkpoints, operation submission, public status and history, artifact inspection and final recommendation commitment. The host validates and executes submitted actions but never chooses a fallback scientific action.

Every operation submission contains a bounded decision audit stating its expected effect, diagnostic target and evidence dependence. Tool receipts retain call order, status, timestamps, error classes and argument/result hashes without retaining raw provider payloads or private chain-of-thought. A provider retry or infrastructure resume is an operational attempt within the same cell, not a new experiment or independent sample. Belief checkpoints are MCP calls inside the existing session rather than separate provider conversations.

9.4 Belief and law-summary checkpoints

Checkpoint records contain prior assessment, predictions, uncertainty, evidence references, an executable law summary, next-experiment intent and overall confidence. The schema permits bounded rationales but not an unconstrained persistent notebook. Predictions are evaluated only after the campaign against sealed truth packs.

9.5 Statistical analysis

The independent units are the 25 task-by-world clusters. The primary contrast is estimated with task fixed effects and a one-sided 0.05 criterion under an intersection–union rule. Three secondary hypotheses use Holm familywise correction. Task-specific contrasts are reported as heterogeneity estimates rather than independent confirmatory claims. Prespecified sensitivity analyses include complete-case, worst-case failed-arm, heteroscedasticity-robust and task-stratified cluster-bootstrap analyses.

9.6 Reproducibility and failure accounting

Participant trajectories, evaluator truth packs and blind-replay packs are stored separately and joined through immutable receipts. Every terminal participant trajectory must pass physical replay, campaign-resource replay and hidden-boundary audit. Provider process attempts, provider sessions, tool calls, operation attempts, committed operations, complete experiments, cells and evaluator executions are reported with distinct denominators.

10. Data and code availability

The executable environment, frozen protocols, analysis code, source data and reproducible figure scripts will accompany the public release. Raw provider payloads, credentials and private world identities are excluded. The private cohort commitment will be released before execution, and the identities and results will be disclosed with the final confirmation package.

11. Author contributions

Author contributions will be finalized before submission using a role-based contribution statement.

12. Competing interests

The authors declare no competing interests.

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